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### CONNECTING PIECE

#### Abstract

A connecting piece includes a first connecting portion, a second connecting portion and an elastic portion. The elastic portion includes a first subsidiary elastic portion and a second subsidiary elastic portion. The first subsidiary elastic portion and the second subsidiary elastic portion are spaced apart from each other and respectively connected between the first connecting portion and the second connecting portion. The first subsidiary elastic portion has a plurality of first grooves. The first grooves are distributed on a side of the first subsidiary elastic portion away from the second subsidiary elastic portion. The second subsidiary elastic portion has a plurality of second grooves. The second grooves are distributed on a side of the second subsidiary elastic portion away from the first subsidiary elastic portion.

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## Background/Summary

### RELATED APPLICATIONS

[0001] This application claims priority to Taiwanese Application Serial Number 113107602, filed Mar. 1, 2024, which is herein incorporated by reference in its entirety.

### BACKGROUND

#### Technical Field

[0002] The present disclosure relates to connecting pieces.

#### Description of Related Art

[0003] As the living standard of people continues to increase nowadays, the importance of electronic products in the daily lives of people also becomes higher and higher. Facing to the huge demand for electronic products, the major manufacturers are also working hard to enhance the market competitiveness of their brands.

[0004] Specifically, in addition to enhancing the functions and performances of electronic products, how to improve the structural quality of electronic products while reducing the production cost is undoubtedly an issue that various manufacturers highly concern.

### SUMMARY

[0005] A technical aspect of the present disclosure is to provide a connecting piece, which can be connected between two objects while flexibly providing a predetermined elasticity.

[0006] According to an embodiment of the present disclosure, a connecting piece includes a first connecting portion, a second connecting portion and an elastic portion. The elastic portion includes a first subsidiary elastic portion and a second subsidiary elastic portion. The first subsidiary elastic portion and the second subsidiary elastic portion are spaced apart from each other and respectively connected between the first connecting portion and the second connecting portion. The first subsidiary elastic portion has a plurality of first grooves. The first grooves are distributed on a side of the first subsidiary elastic portion away from the second subsidiary elastic portion. The second subsidiary elastic portion has a plurality of second grooves. The second grooves are distributed on a side of the second subsidiary elastic portion away from the first subsidiary elastic portion.

[0007] In one or more embodiments of the present disclosure, the first grooves align with the second grooves.

[0008] In one or more embodiments of the present disclosure, a quantity of the first grooves is equal to a quantity of the second grooves.

[0009] In one or more embodiments of the present disclosure, the first subsidiary elastic portion has a plurality of third grooves and a plurality of fourth grooves. The third grooves and the fourth grooves are distributed on two opposite sides of the first subsidiary elastic portion. Each of the third grooves is located between adjacent two of the first grooves. Each of the fourth grooves is located between adjacent two of the first grooves.

[0010] In one or more embodiments of the present disclosure, the third grooves align with the fourth grooves.

[0011] In one or more embodiments of the present disclosure, a quantity of the third grooves is equal to a quantity of the fourth grooves.

[0012] In one or more embodiments of the present disclosure, the second subsidiary elastic portion has a plurality of fifth grooves and a plurality of sixth grooves. The fifth grooves and the sixth grooves are distributed on two opposite sides of the second subsidiary elastic portion. The fifth grooves and the sixth grooves are staggered from each other. Each of the fifth grooves is located between adjacent two of the second grooves. Each of the sixth grooves is located between adjacent

two of the second grooves.

[0013] In one or more embodiments of the present disclosure, each of the fifth grooves aligns with a corresponding one of the third grooves. Each of the sixth grooves aligns with a corresponding one of the fourth grooves.

[0014] In one or more embodiments of the present disclosure, the connecting piece further includes a positioning portion. The positioning portion is connected between the first connecting portion and the elastic portion. The first connecting portion has a first outer diameter. The elastic portion has a second outer diameter. The positioning portion has a third outer diameter. The third outer diameter is larger than the first outer diameter and the second outer diameter.

[0015] In one or more embodiments of the present disclosure, the first connecting portion has a first thread. The second connecting portion has a second thread.

[0016] In one or more embodiments of the present disclosure, the first thread is an inner thread. The second connecting portion has a through hole. The connecting piece further includes a supporting rod. The supporting rod includes a rod body and a threaded portion connected with the rod body. The threaded portion is coupled with the first thread. The rod body is at least partially located in the through hole. The first subsidiary elastic portion and the second subsidiary elastic portion abut against two opposite sides of the rod body.

[0017] According to an embodiment of the present disclosure, a connecting piece includes a first connecting portion, a second connecting portion and an elastic portion. The elastic portion extends along a spiral direction to form a spiral shape. The elastic portion is connected between the first connecting portion and the second connecting portion. The elastic portion has a cross-sectional area. The cross-sectional area is perpendicular to the spiral direction and has a rectangular shape.

[0018] In one or more embodiments of the present disclosure, the first connecting portion has a first thread. The second connecting portion has a second thread.

[0019] In one or more embodiments of the present disclosure, the first thread is an inner thread. The second connecting portion has a through hole. The connecting piece further includes a supporting rod. The supporting rod includes a rod body and a threaded portion connected with the rod body. The threaded portion is coupled with the first thread. The rod body is at least partially located in the through hole and surrounded by the elastic portion. The elastic portion abuts against the rod body.

[0020] In one or more embodiments of the present disclosure, the first connecting portion, the second connecting portion and the elastic portion are of an integrally formed structure.

[0021] The above-mentioned embodiments of the present disclosure have at least the following advantages: [0022] (1) Since the first connecting portion, the second connecting portion, the elastic portion and the positioning portion are of an integrally formed structure, apart from reducing the production cost of the connecting piece, the structural strength of the connecting piece is also effectively increased. [0023] (2) During the designing process, by changing the quantity and the size of the first grooves, the third grooves and the fourth grooves on the first subsidiary elastic portion, the elasticity of the first subsidiary elastic portion can be adjusted in a simple and easy manner. By changing the quantity and the size of the second grooves, the fifth grooves and the sixth grooves on the second subsidiary elastic portion, the elasticity of the second subsidiary elastic portion can be adjusted in a simple and easy manner. Thus, a user can design the overall elasticity of the elastic portion in a simple and easy manner. [0024] (3) Provided that the first subsidiary elastic portion and the second subsidiary elastic portion are not structurally symmetrical, one of the first subsidiary elastic portion and the second subsidiary elastic portion can achieve a larger elastic deformation than the other when being compressed. Thus, the connecting piece can achieve an appropriate amount of curvature, which can enhance the application flexibility of the connecting piece.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

[0026] FIG. 1 is a schematic view of a connecting piece according to an embodiment of the present disclosure;

[0027] FIG. 2 is a schematic view of the connecting piece of FIG. 1 at another visual angle;

[0028] FIG. 3 is a front view of the connecting piece of FIGS. 1-2 along the arrow SA;

[0029] FIG. 4 is a sectional view along the sectional line M-M of FIG. 3;

[0030] FIG. 5 is a front view of the connecting piece of FIGS. 1-2 along the arrow SB;

[0031] FIG. 6 is a front view of the connecting piece of FIGS. 1-2 along the arrow SC;

[0032] FIG. 7 is a front view of the connecting piece of FIGS. 1-2 along the arrow SD;

[0033] FIG. 8 is a sectional view of a connecting piece according to another embodiment of the present disclosure;

[0034] FIG. 9 is a front view of a connecting piece according to a further embodiment of the present disclosure;

[0035] FIG. 10 is a sectional view of the connecting piece of FIG. 9; and

[0036] FIG. 11 is a sectional view of a connecting piece according to another embodiment of the present disclosure.

### DETAILED DESCRIPTION

[0037] Drawings will be used below to disclose embodiments of the present disclosure. For the sake of clear illustration, many practical details will be explained together in the description below. However, it is appreciated that the practical details should not be used to limit the claimed scope. In other words, in some embodiments of the present disclosure, the practical details are not essential. Moreover, for the sake of drawing simplification, some customary structures and elements in the drawings will be schematically shown in a simplified way. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

[0038] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meanings as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0039] Reference is made to FIGS. 1-2. FIG. 1 is a schematic view of a connecting piece **100** according to an embodiment of the present disclosure. FIG. 2 is a schematic view of the connecting piece **100** of FIG. 1 at another visual angle. In this embodiment, as shown in FIGS. 1-2, a connecting piece **100** includes a first connecting portion **110**, a second connecting portion **120** and an elastic portion **130**. The elastic portion **130** includes a first subsidiary elastic portion **131** and a second subsidiary elastic portion **132**. The first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** are spaced apart from each other. The first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** respectively extend along a first direction **D1** and provide a certain amount of elasticity. Moreover, the first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** are respectively connected between the first connecting portion **110** and the second connecting portion **120** along the first direction **D1**. Therefore, the distance between the first connecting portion **110** and the second connecting portion **120** can be elastically changed. This means the first connecting portion **110** and the second connecting portion **120** can be moved close to or away from each other along the first direction **D1**. The first connecting portion **110** is configured to connect with a first connecting object **200**. The second

connecting portion **120** is configured to connect with a second connecting object **300** (for the sake of drawing simplification, the first connecting object **200** and the second connecting object **300** are only shown in FIG. 3). In simple words, the connecting piece **100** can be elastically connected between the first connecting object **200** and the second connecting object **300**. For example, the first connecting object **200** and the second connecting object **300** can be but are not limited to modules such as heat sink or back panel.

[0040] To be more specific, in this embodiment, the connecting piece **100** further includes a positioning portion **140**. As shown in FIGS. 1-2, the positioning portion **140** is connected between the first connecting portion **110** and the elastic portion **130** along the first direction D1. The first connecting portion **110** has a first outer diameter X1. The elastic portion **130** has a second outer diameter X2. The positioning portion **140** has a third outer diameter X3. The third outer diameter X3 is larger than the first outer diameter X1 and the second outer diameter X2, which is beneficial for the positioning portion **140** to abut against the first connecting object **200**.

[0041] Moreover, for example, as shown in FIGS. 1-2, the first connecting portion **110** has a first thread **110s**, and the first thread **110s** is an inner thread. In other embodiments, the first thread **110s** can be an outer thread. On the other hand, according to the actual situation, the first connecting portion **110** can also be designed to be a rivet or a tenon, for the sake of connection with the first connecting object **200**.

[0042] Furthermore, for example, as shown in FIGS. 1-2, the second connecting portion **120** has a second thread **120s**, and the second thread **120s** is an outer thread. In other embodiments, the second thread **120s** can be an inner thread. Similarly, according to the actual situation, the second connecting portion **120** can also be designed to be a rivet or a tenon, for the sake of connection with the second connecting object **300**.

[0043] In practical applications, the first connecting portion **110**, the second connecting portion **120**, the elastic portion **130** and the positioning portion **140** are of an integrally formed structure. This means the first connecting portion **110**, the second connecting portion **120**, the elastic portion **130** and the positioning portion **140** are processed from the same piece of material. Accordingly, apart from reducing the production cost of the connecting piece **100**, the structural strength of the connecting piece **100** is also effectively increased.

[0044] Reference is made to FIG. 3. FIG. 3 is a front view of the connecting piece **100** of FIGS. 1-2 along the arrow SA. As shown in FIG. 3, the connecting piece **100** is connected between the first connecting object **200** and the second connecting object **300** along the first direction D1, while the positioning portion **140** abuts against the first connecting object **200**, which is beneficial to fix the relative positions between the connecting piece **100** and the first connecting object **200**. In addition, in this embodiment, as shown in FIGS. 1-3, the first subsidiary elastic portion **131** has a plurality of first grooves G1. The first grooves G1 are distributed along the first direction D1 on a side of the first subsidiary elastic portion **131** away from the second subsidiary elastic portion **132**. The second subsidiary elastic portion **132** has a plurality of second grooves G2. The second grooves G2 are distributed along the first direction D1 on a side of the second subsidiary elastic portion **132** away from the first subsidiary elastic portion **131**. In other words, the first grooves G1 and the second grooves G2 are located on two opposite sides of the elastic portion **130** along a second direction D2, in which the second direction D2 is perpendicular to the first direction D1.

[0045] Furthermore, in this embodiment, as shown in FIGS. 1-3, a quantity of the first grooves G1 is equal to a quantity of the second grooves G2, while the first grooves G1 align with the second grooves G2. This means each of the first grooves G1 and a corresponding one of the second grooves G2 are located at the same level in the first direction D1.

[0046] Reference is made to FIG. 4. FIG. 4 is a sectional view along the sectional line M-M of FIG. 3. In this embodiment, as shown in FIG. 4, a cross-section of the elastic portion **130** perpendicular to the first direction D1 has a circular shape, while an outer edge **131e** of the first subsidiary elastic portion **131** and an outer edge **132e** of the second subsidiary elastic portion **132**

define a perimeter of this circular shape. In practice, the elastic portion **130** with a circular cross-section can be beneficial to the manufacturing process of the connecting piece **100**, and also improve the structural strength of the connecting piece **100**. However, in other embodiments, according to the actual situation, the cross-section of the elastic portion **130** perpendicular to the first direction **D1** can be non-circular, such as square or polygonal. However, this does not intend to limit the present disclosure.

[0047] Reference is made to FIG. 5. FIG. 5 is a front view of the connecting piece **100** of FIGS. 1-2 along the arrow SB. In this embodiment, as shown in FIG. 5, the first subsidiary elastic portion **131** has a plurality of third grooves **G3** and a plurality of fourth grooves **G4**. The third grooves **G3** and the fourth grooves **G4** are distributed on two opposite sides of the first subsidiary elastic portion **131** along the first direction **D1**. Each of the third grooves **G3** is located between adjacent two of the first grooves **G1**. Each of the fourth grooves **G4** is located between adjacent two of the first grooves **G1**. In other words, the third grooves **G3** and the fourth grooves **G4** are located on two opposite sides of the first subsidiary elastic portion **131** along a third direction **D3**, in which the third direction **D3** is perpendicular to the first direction **D1** and the second direction **D2**.

[0048] Furthermore, in this embodiment, as shown in FIG. 5, a quantity of the third grooves **G3** is equal to a quantity of the fourth grooves **G4**, while the third grooves **G3** align with the fourth grooves **G4**. This means each of the third grooves **G3** and a corresponding one of the fourth grooves **G4** are located at the same level in the first direction **D1**.

[0049] During the designing process, by changing the quantity and the size of the first grooves **G1**, the third grooves **G3** and the fourth grooves **G4** on the first subsidiary elastic portion **131**, the elasticity of the first subsidiary elastic portion **131** can be adjusted in a simple and easy manner.

[0050] Reference is made to FIGS. 6-7. FIG. 6 is a front view of the connecting piece **100** of FIGS. 1-2 along the arrow SC. FIG. 7 is a front view of the connecting piece **100** of FIGS. 1-2 along the arrow SD. In this embodiment, as shown in FIGS. 6-7, the second subsidiary elastic portion **132** has a plurality of fifth grooves **G5** and a plurality of sixth grooves **G6**. The fifth grooves **G5** and the sixth grooves **G6** are distributed on two opposite sides of the second subsidiary elastic portion **132** along the first direction **D1**. The fifth grooves **G5** and the sixth grooves **G6** are staggered from each other. This means the fifth grooves **G5** and the sixth grooves **G6** are not located at the same level in the first direction **D1**. Each of the fifth grooves **G5** is located between adjacent two of the second grooves **G2**. Each of the sixth grooves **G6** is located between adjacent two of the second grooves **G2**.

[0051] In addition, as shown in FIG. 7, each of the fifth grooves **G5** aligns with a corresponding one of the third grooves **G3**. Moreover, as shown in FIG. 3, each of the sixth grooves **G6** aligns with a corresponding one of the fourth grooves **G4**. Since the fifth grooves **G5** and the sixth grooves **G6** are staggered from each other, as shown in FIG. 5, the second subsidiary elastic portion **132** can overlap with at least one of the third grooves **G3** and the fourth grooves **G4** in the second direction **D2**.

[0052] Similarly, during the designing process, by changing the quantity and the size of the second grooves **G2**, the fifth grooves **G5** and the sixth grooves **G6** on the second subsidiary elastic portion **132**, the elasticity of the second subsidiary elastic portion **132** can be adjusted in a simple and easy manner.

[0053] It is worth to note that, in this embodiment, as shown in FIGS. 1-3 and 5-7, the first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** are not structurally symmetrical. Therefore, the first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** can provide different magnitudes of elasticity, such that a user can design the overall elasticity of the connecting piece **100** in a more flexible manner.

[0054] Furthermore, provided that the first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** are not structurally symmetrical, one of the first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** can achieve a larger elastic deformation than the other

when being compressed. Thus, the connecting piece **100** can achieve an appropriate amount of curvature, which can enhance the application flexibility of the connecting piece **100**.

[0055] Reference is made to FIG. **8**. FIG. **8** is a sectional view of a connecting piece **100** according to another embodiment of the present disclosure. In this embodiment, as shown in FIG. **8**, the first thread **110s** of the first connecting portion **110** is an inner thread. The second connecting portion **120** has a through hole H. The connecting piece **100** further includes a supporting rod **150**. The supporting rod **150** includes a rod body **151** and a threaded portion **152** connected with the rod body **151**. The threaded portion **152** of the supporting rod **150** is coupled with the first thread **110s** of the first connecting portion **110**. The rod body **151** of the supporting rod **150** is at least partially located in the through hole H. The first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** abut against two opposite sides of the rod body **151**. In this way, the first subsidiary elastic portion **131** and the second subsidiary elastic portion **132** are unlikely to be bent, such that the overall strength of the connecting piece **100** is effectively increased.

[0056] Reference is made to FIG. **9**. FIG. **9** is a front view of a connecting piece **100** according to a further embodiment of the present disclosure. As shown in FIG. **9**, a connecting piece **100** includes a first connecting portion **110**, a second connecting portion **120** and an elastic portion **130**. In this embodiment, the elastic portion **130** extends along a spiral direction DS to form a spiral shape. The elastic portion **130** is connected between the first connecting portion **110** and the second connecting portion **120**.

[0057] Reference is made to FIG. **10**. FIG. **10** is a sectional view of the connecting piece **100** of FIG. **9**. In this embodiment, as shown in FIG. **10**, the elastic portion **130** has a cross-sectional area CA. The cross-sectional area CA is perpendicular to the spiral direction DS and has a rectangular shape. In this way, the elastic portion **130** is appropriate to be made and processed by machines. Moreover, by changing the dimension and size of the cross-sectional area CA, the elasticity of the elastic portion **130** can be designed in a simple and easy manner. In practical applications, the first connecting portion **110**, the second connecting portion **120** and the elastic portion **130** are of an integrally formed structure. This means the connecting piece **100** is processed from the same piece of material. Accordingly, apart from reducing the production cost of the connecting piece **100**, the structural strength of the connecting piece **100** is also effectively increased.

[0058] Furthermore, as shown in FIGS. **9-10**, the first connecting portion **110** has a first thread **110s**. The second connecting portion **120** has a second thread **120s**. On the other hand, according to the actual situation, the first connecting portion **110** and the second connecting portion **120** can also be designed to be a rivet or a tenon, for the sake of respective connection with the first connecting object **200** and the second connecting object **300** (please refer to FIG. **3** for the first connecting object **200** and the second connecting object **300**).

[0059] Reference is made to FIG. **11**. FIG. **11** is a sectional view of a connecting piece **100** according to another embodiment of the present disclosure. In this embodiment, as shown in FIG. **11**, the first thread **110s** of the first connecting portion **110** is an inner thread. The second connecting portion **120** has a through hole H. The connecting piece **100** further includes a supporting rod **150**. The supporting rod **150** includes a rod body **151** and a threaded portion **152** connected with the rod body **151**. The threaded portion **152** of the supporting rod **150** is coupled with the first thread **110s** of the first connecting portion **110**. The rod body **151** of the supporting rod **150** is at least partially located in the through hole H and surrounded by the spirally-shaped elastic portion **130**. The elastic portion **130** abuts against the rod body **151**. In this way, the elastic portion **130** is unlikely to be bent, such that the overall strength of the connecting piece **100** is effectively increased.

[0060] In conclusion, the aforementioned embodiments of the present disclosure have at least the following advantages: [0061] (1) Since the first connecting portion, the second connecting portion, the elastic portion and the positioning portion are of an integrally formed structure, apart from reducing the production cost of the connecting piece, the structural strength of the connecting piece

is also effectively increased. [0062] (2) During the designing process, by changing the quantity and the size of the first grooves, the third grooves and the fourth grooves on the first subsidiary elastic portion, the elasticity of the first subsidiary elastic portion can be adjusted in a simple and easy manner. By changing the quantity and the size of the second grooves, the fifth grooves and the sixth grooves on the second subsidiary elastic portion, the elasticity of the second subsidiary elastic portion can be adjusted in a simple and easy manner. Thus, a user can design the overall elasticity of the elastic portion in a simple and easy manner. [0063] (3) Provided that the first subsidiary elastic portion and the second subsidiary elastic portion are not structurally symmetrical, one of the first subsidiary elastic portion and the second subsidiary elastic portion can achieve a larger elastic deformation than the other when being compressed. Thus, the connecting piece can achieve an appropriate amount of curvature, which can enhance the application flexibility of the connecting piece.

[0064] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0065] It will be apparent to the person having ordinary skill in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of the present disclosure provided they fall within the scope of the following claims.

## Claims

1. A connecting piece, comprising: a first connecting portion; a second connecting portion; and an elastic portion comprising a first subsidiary elastic portion and a second subsidiary elastic portion, the first subsidiary elastic portion and the second subsidiary elastic portion being spaced apart from each other and respectively connected between the first connecting portion and the second connecting portion, the first subsidiary elastic portion having a plurality of first grooves, the first grooves being distributed on a side of the first subsidiary elastic portion away from the second subsidiary elastic portion, the second subsidiary elastic portion having a plurality of second grooves, the second grooves being distributed on a side of the second subsidiary elastic portion away from the first subsidiary elastic portion.
2. The connecting piece of claim 1, wherein the first grooves align with the second grooves.
3. The connecting piece of claim 1, wherein a quantity of the first grooves is equal to a quantity of the second grooves.
4. The connecting piece of claim 1, wherein the first subsidiary elastic portion has a plurality of third grooves and a plurality of fourth grooves, the third grooves and the fourth grooves are distributed on two opposite sides of the first subsidiary elastic portion, each of the third grooves is located between adjacent two of the first grooves, each of the fourth grooves is located between adjacent two of the first grooves.
5. The connecting piece of claim 4, wherein the third grooves align with the fourth grooves.
6. The connecting piece of claim 4, wherein a quantity of the third grooves is equal to a quantity of the fourth grooves.
7. The connecting piece of claim 4, wherein the second subsidiary elastic portion has a plurality of fifth grooves and a plurality of sixth grooves, the fifth grooves and the sixth grooves are distributed on two opposite sides of the second subsidiary elastic portion, the fifth grooves and the sixth grooves are staggered from each other, each of the fifth grooves is located between adjacent two of the second grooves, each of the sixth grooves is located between adjacent two of the second grooves.
8. The connecting piece of claim 7, wherein each of the fifth grooves aligns with a corresponding



one of the third grooves, each of the sixth grooves aligns with a corresponding one of the fourth grooves.

**9.** The connecting piece of claim 1, further comprising: a positioning portion connected between the first connecting portion and the elastic portion, the first connecting portion having a first outer diameter, the elastic portion having a second outer diameter, the positioning portion having a third outer diameter, the third outer diameter being larger than the first outer diameter and the second outer diameter.

**10.** The connecting piece of claim 1, wherein the first connecting portion has a first thread, the second connecting portion has a second thread.

**11.** The connecting piece of claim 10, wherein the first thread is an inner thread, the second connecting portion has a through hole, the connecting piece further comprises: a supporting rod comprising a rod body and a threaded portion connected with the rod body, the threaded portion is coupled with the first thread, the rod body is at least partially located in the through hole, the first subsidiary elastic portion and the second subsidiary elastic portion abut against two opposite sides of the rod body.

**12.** A connecting piece, comprising: a first connecting portion; a second connecting portion; and an elastic portion extending along a spiral direction to form a spiral shape, the elastic portion being connected between the first connecting portion and the second connecting portion, the elastic portion having a cross-sectional area, the cross-sectional area being perpendicular to the spiral direction and having a rectangular shape.

**13.** The connecting piece of claim 12, wherein the first connecting portion has a first thread, the second connecting portion has a second thread.

**14.** The connecting piece of claim 13, wherein the first thread is an inner thread, the second connecting portion has a through hole, the connecting piece further comprises: a supporting rod comprising a rod body and a threaded portion connected with the rod body, the threaded portion is coupled with the first thread, the rod body is at least partially located in the through hole and surrounded by the elastic portion, the elastic portion abuts against the rod body.

**15.** The connecting piece of claim 12, wherein the first connecting portion, the second connecting portion and the elastic portion are of an integrally formed structure.

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